



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
08-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

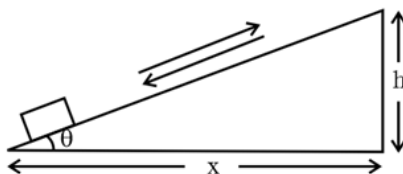
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

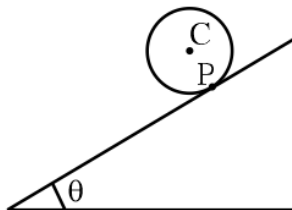
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Block of mass 2kg is placed on an inclined plane on which it does not spontaneously start to slide. It is pulled up, and then down the incline to the initial position very slowly in such a manner that the applied force is always parallel to the slope. Work done by external agent is 40 J. Assume $\mu_s \simeq \mu_k = \mu$.



- (A) The maximum possible value of h is 0.5 m
- (B) The maximum possible value of h is 1 m
- (C) For a given value of total work done by external agent, the length of incline's base depends on the coefficient of friction
- (D) For a given value of total work done by external agent, the length of the incline's base does not depend upon the angle of incline plane

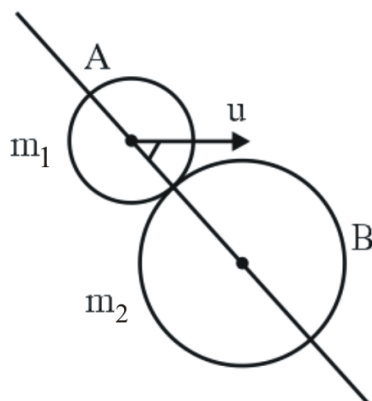
2. A solid cylinder is released on an inclined plane of inclination θ where $\sin \theta = \frac{3}{5}$. The coefficient of friction between the cylinder and plane is $\mu = 0.2$. [Radius of cylinder is $\frac{10}{3}$ cm and mass is 1 kg]



- (A) The time after which acceleration of point of contact 'P' makes 45° angle with velocity of center 'C' is $\frac{1}{16}$ sec
- (B) Work done by friction in ground frame on the cylinder in 1 second is -1.76 J
- (C) Work done by friction in frame of center 'C' on the cylinder in $\frac{1}{16}$ second is 0.01 J
- (D) Work done by friction in ground frame is negative but positive in frame of center 'C' of cylinder

Space for Rough Work

3. In an oblique collision between two spheres of mass m_1 and m_2 , sphere of mass m_1 is moving with a certain speed 'u' towards sphere of mass m_2 at rest. The velocity vector of moving sphere makes angle α with the common normal at the point of impact.

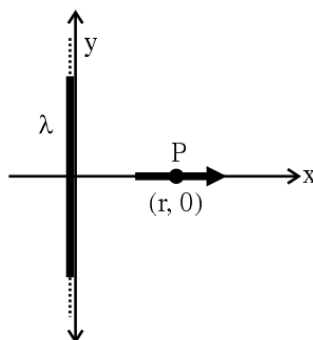


After a perfectly elastic collision the velocity of first sphere makes angle θ with the direction it was initially moving. For different impact parameters the angle α changes.

- (A) The speed of first sphere in the frame of centre of mass of m_1 and m_2 is $\frac{m_2 u}{m_1 + m_2}$ after collision
- (B) For $m_2 > m_1$, angle θ can have any value
- (C) $\tan \theta = \frac{\tan \alpha (1 - M)}{\tan^2 \alpha + M}$, where $M = \frac{m_1 - m_2}{m_1 + m_2}$
- (D) For $m_2 < m_1$, the maximum value of angle θ is $\sin^{-1} \left(\frac{m_2}{m_1} \right)$

Space for Rough Work

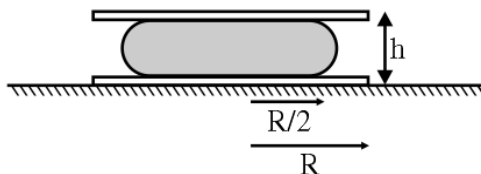
4. An infinite wire of linear charge density ' λ ' is placed along y-axis and a dipole $\vec{p}$ is placed along x-axis at position $(r, 0)$ as shown in figure. Choose the **CORRECT** statement (s)



- (A) Electric field intensity will be zero on line $x = r$ in x-y plane at $\left(r, \left(\frac{pr}{2\lambda}\right)^{\frac{1}{3}}\right)$
- (B) If dipole is rotated anticlockwise in x-y plane by 90° then electric field intensity will be zero at a distance r from dipole if $\frac{p}{\lambda} = \sqrt{6} \left(\frac{\sqrt{3} + 1}{2}\right) r^2$
- (C) Force on dipole in given orientation is $\frac{-\lambda p}{2\pi\epsilon_0 r^2} \hat{i}$
- (D) If dipole is rotated then, minimum force on dipole can be zero for a certain orientation.

Space for Rough Work

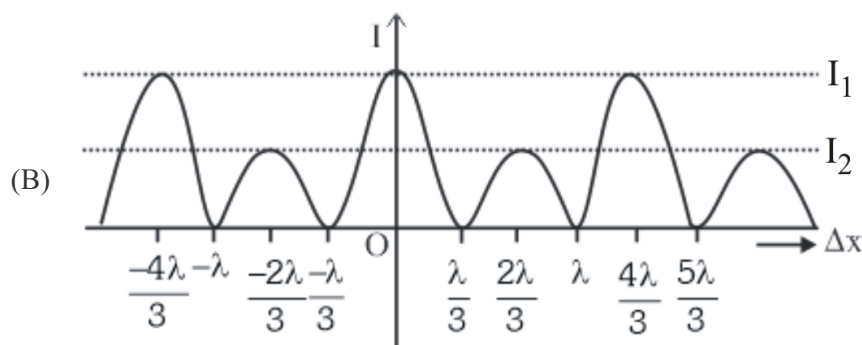
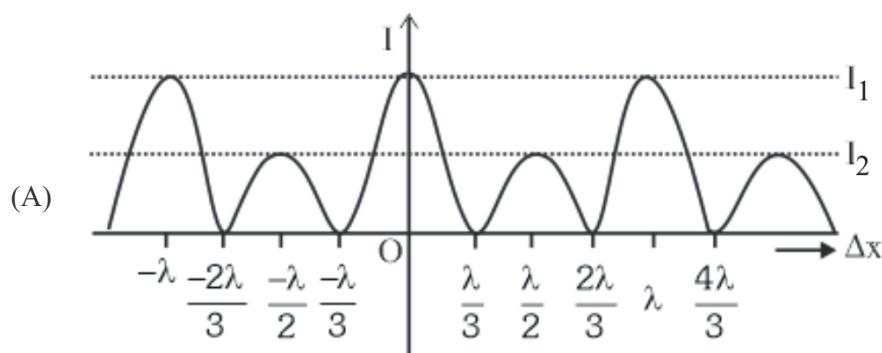
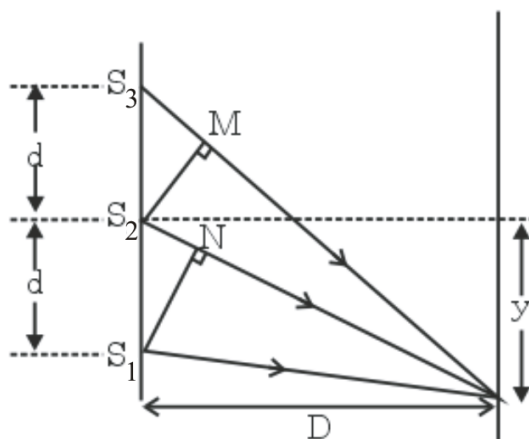
5. A container is completely filled with a liquid. Coefficient of linear expansion of container is α and coefficient of volume expansion of liquid is γ . What must be the condition so that liquid does not flow out on changing the temperature ?
- (A) $3\alpha = \gamma$ (B) $3\alpha \geq \gamma$
 (C) $3\alpha > \gamma$ (D) $3\alpha \leq \gamma$
6. A liquid pellet when placed between two circular plates of radius R takes the form of a circular disc of radius $\frac{R}{2}$. Angle of contact of liquid is 120° . Mass of the upper plate is 18 g. How much extra mass (in g) should be placed on upper plate so that liquid is about to extrude from the plates? ($R \gg h$)



- (A) 558 (B) 270 (C) 126 (D) 18

Space for Rough Work

7. A triple slit experiment is performed as shown in the figure with monochromatic light of amplitude A_0 . $S_3M = S_2N = \Delta x$. The graph between resultant intensity k versus Δx is plotted choose the correct option(s). ($d \ll D, y \ll D$)



- (C) $\frac{I_1 + I_2}{I_1 - I_2} = \frac{5}{4}$, where I_1 and I_2 are the shown in above options
- (D) $\frac{I_1 + I_2}{I_1 - I_2} = 5$, where I_1 and I_2 are the shown in above options

Space for Rough Work

8. A single charged particle of mass 4 times the mass of electron and charge ' $-2e$ ' is revolving in circular path around stationary nucleus of helium. (Consider that Bohr model is applicable). Select the **CORRECT** option(s) :
- (A) Radius of ground state is $0.033 \text{ \AA}$
- (B) Velocity of particle is $\left(\frac{2e^2}{\epsilon_0 h} \right)$ in $n = 1$
- (C) Energy of series limit of 1^{st} excited state is 217.6 eV
- (D) Energy of series limit of 1^{st} excited state is 54.4 eV
-

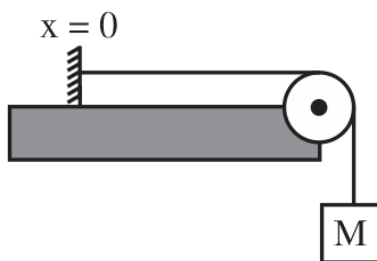
Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. In a sonometer wire experiment, length of string on table is ℓ . Take fixed end as $x = 0$. String is vibrating in first overtone. It is in unison with tuning fork of frequency f_0 . At $t = 0$, all particles are at mean position and particles in vicinity of $x = 0$ are moving in upward direction. In list-I location on string is provided. In list-II information about instantaneous power at the locations provided in list-I is given. (Consider flow of energy to be positive towards right)



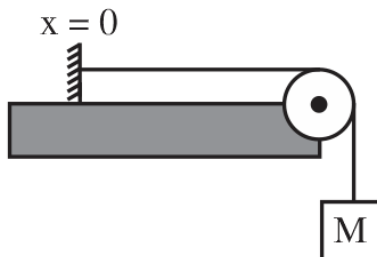
List-I		List-II	
(I)	$x = \frac{\ell}{2}$	(P)	Zero
(II)	$x = \frac{\ell}{4}$	(Q)	Maximum
(III)	$x = \frac{\ell}{3}$	(R)	Half of maximum
(IV)	$x = \frac{\ell}{6}$	(S)	$\frac{\sqrt{3}}{2}$ times of maximum
		(T)	Positive
		(U)	Negative

Which of the following is **INCORRECT** at $t = \frac{1}{8f_0}$

- (A) I $\rightarrow$ P (B) II $\rightarrow$ P (C) III $\rightarrow$ R,U (D) IV $\rightarrow$ R,U

Space for Rough Work

10. In a sonometer wire experiment, length of string on table is ℓ . Take fixed end as $x = 0$. String is vibrating in first overtone. It is in unison with tuning fork of frequency f_0 . At $t = 0$, all particles are at mean position and particles in vicinity of $x = 0$ are moving in upward direction. In list-I location on string is provided. In list-II information about instantaneous power at the locations provided in list-I is given. (Consider flow of energy to be positive towards right)



List-I		List-II	
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(III)	$x = \frac{\ell}{3}$	(R)	Half of maximum
(IV)	$x = \frac{\ell}{6}$	(S)	$\frac{\sqrt{3}}{2}$ times of maximum
		(T)	Positive
		(U)	Negative

Now mass m is submerged in a liquid having density $\frac{5}{9}$ times that of object. Which of the following is **CORRECT** at $t = \frac{1}{4f_0}$?

- (A) $I \rightarrow P$ (B) $II \rightarrow Q$ (C) $III \rightarrow S$ (D) $IV \rightarrow T$

Space for Rough Work

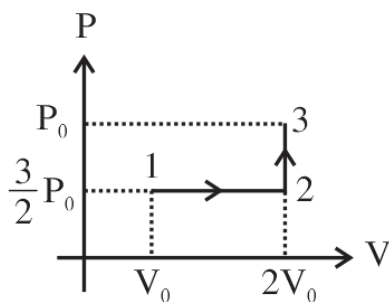
11. In thermodynamic process on an ideal monoatomic gas. The infinitesimal heat absorbed by the gas is given by $T\Delta x$, here T = temperature of system, Δx = infinitesimal change in thermodynamic quantity x of the system for a mole of monatomic ideal gas.

$$x = \frac{3}{2}R \ln\left(\frac{T}{T_A}\right) + R \ln\left(\frac{V}{V_A}\right) \text{ here } R = \text{gas constant } V = \text{volume of gas, } (T_A, V_A = \text{constant}).$$

List-I below gives some quantities involved in a thermodynamic process defined by states 1, 2 and 3 and List-II gives some possible values of quantities.

List-I		List-II	
(I)	Work done by system in process $1 \rightarrow 2 \rightarrow 3$	(P)	$\frac{1}{3}RT_0 \ln 2$
(II)	Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$	(Q)	$\frac{1}{3}RT_0$
(III)	Heat absorbed by system in process $1 \rightarrow 2 \rightarrow 3$	(R)	RT_0
(IV)	Heat absorbed by system in process $1 \rightarrow 2$	(S)	$\frac{4}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3 + \ln 2)$
		(U)	$\frac{5}{6}RT_0$

If the process carried out on one mole of monoatomic ideal gas as shown in figure in PV curve with $P_0V_0 = \frac{1}{3}RT_0$. The correct match is :



- (A) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ U
 (B) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ S; IV $\rightarrow$ U
 (C) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ P; IV $\rightarrow$ U
 (D) I $\rightarrow$ S; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ T

Space for Rough Work

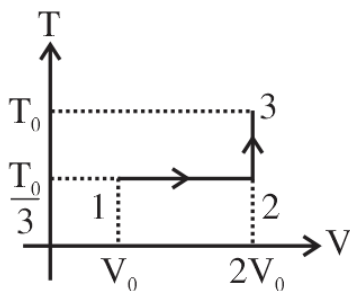
12. In thermodynamic process on an ideal monoatomic gas. The infinitesimal heat absorbed by the gas is given by $T\Delta x$, here T = temperature of system, Δx = infinitesimal change in thermodynamic quantity x of the system for a mole of monatomic ideal gas.

$$x = \frac{3}{2}R \ln\left(\frac{T}{T_A}\right) + R \ln\left(\frac{V}{V_A}\right) \text{ here } R = \text{gas constant } V = \text{volume of gas, } (T_A, V_A = \text{constant}).$$

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(III)	Heat absorbed by system in process $1 \rightarrow 2 \rightarrow 3$	(R)	RT_0
(IV)	Heat absorbed by system in process $1 \rightarrow 2$	(S)	$\frac{4}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3 + \ln 2)$
		(U)	$\frac{5}{6}RT_0$

If the process is carried out on the mole of monoatomic ideal gas as shown in figure in TV curve with $P_0V_0 = \frac{1}{3}RT_0$. The correct match is



- (A) I $\rightarrow$ P; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ T
 (B) I $\rightarrow$ P; II $\rightarrow$ T; III $\rightarrow$ Q; IV $\rightarrow$ U
 (C) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ T; IV $\rightarrow$ U
 (D) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ T; IV $\rightarrow$ P

Space for Rough Work

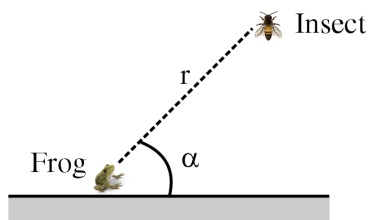
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

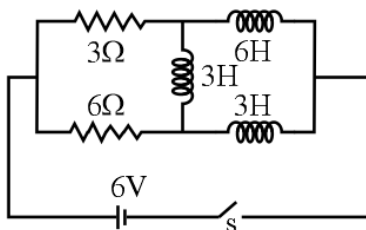
Zero Marks : 0 In all other cases.

1. In the given figure, a frog sitting on leaf on the surface of water spots an insect on a tree branch. The shortest distance between frog and insect is ' $r = 0.1$ m' and the line joining makes an angle ' α ' as shown, such that ' $\alpha = \pi/6$ radians'. Find the square of minimum velocity with which frog launches itself in order to eat the insect. ($g = 10 \text{ m/s}^2$).

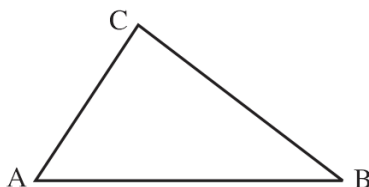


Space for Rough Work

2. An ideal battery of 6V EMF is connected to the given circuit as shown. All inductors are ideal. Initially no current is flowing through inductors. What will be current (in A) through inductor of inductance 6H at steady state?

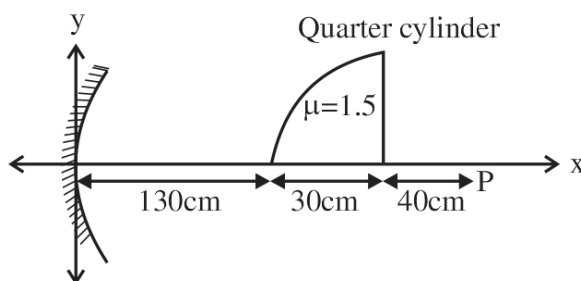


3. A right angle prism ABC of refractive index 2 is shown in figure. Two of its acute angles are 30° and 60° . It is surrounded by a medium having a variable refractive index 'n' varying with time t in seconds. A ray of light is incident normally on the face AB. At $t = 0$ the ray undergoes total internal reflection at one of the faces and grazes on the surface of the other face. The emergent ray from the prism rotates at a constant rate $\frac{\pi}{6}$ rad/s. Find the value of square of refractive index 'n' at $t = 2$ sec. (take $\sqrt{3} = 1.73$)



Space for Rough Work

4. If minimum energy that an electron has is 200 KeV, then the width that may be resolved using an electron microscope is d_1 . Now if a light microscope is used such that photon has same energy as that of electron, the width resolved is d_2 . Find d_1/d_2 ?
5. As a result of α -decay to gain stability a heavy nucleus X of mass number 204 disintegrates and daughter nuclei is formed which is at an excited state. Daughter nuclei then de-excites by emitting a γ photon such that its kinetic energy increases in same direction. Assuming that nearly all the energy released during γ photon emission is taken away by the γ -photon, the change in kinetic energy of the daughter nuclei is 'k' keV. Find k.
[BE of X = 1480 MeV, BE of Y^* = 1505.5 MeV, BE of Y = 1507.5 MeV, Take mass of nucleon = 1.6×10^{-27} kg]
BE of α particle = 28.3 MeV
6. A point object is kept at a distance of 2m from origin on x axis. A quarter cylinder of radius $R = 30$ cm and refractive index $\mu = 1.5$ having its axis perpendicular to the plane of paper is placed between the object and the parabolic reflector at a distance 130 cm from origin as shown. What will be the position of image (in cm) after reflection from parabolic reflector $y^2 = 2x$ (Here x and y are in meter)



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. The value of four quantum numbers for the last electron of atom of element 'X' are $n = 7$, $l = 1$, $m = +1$ and $s = +\frac{1}{2}$ or $-\frac{1}{2}$ and value of spin only magnetic moment for element 'X' is zero. Element 'X' has two isotopes (I) A_ZX and, (II) B_ZX . (Given ' $B - A = B - 2Z = 18$ ', where A and B are atomic masses and Z is atomic number). The possible value of all four quantum numbers for 90th electron of atom of element 'X' is

(A) $n = 6, l = 0, m = 0, s = -\frac{1}{2}$

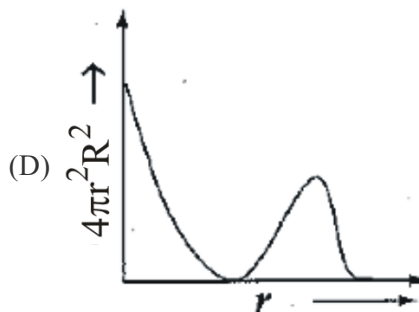
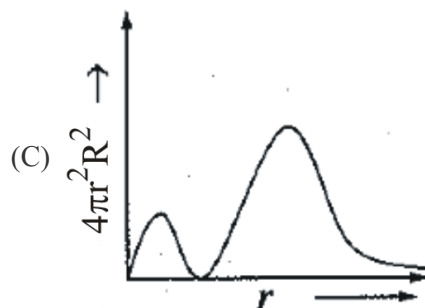
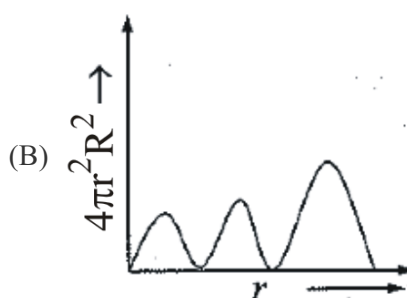
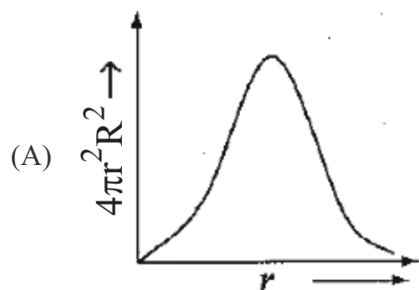
(B) $n = 5, l = 2, m = -1, s = +\frac{1}{2}$

(C) $n = 6, l = 0, m = 0, s = +\frac{1}{2}$

(D) $n = 5, l = 3, m = -2, s = -\frac{1}{2}$

Space for Rough Work

2. The **correct** information related with 0.09 M solution of $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$, ethylenediamine (en) is ($K_{b1} = 8.1 \times 10^{-5}$, $K_{b2} = 7.0 \times 10^{-8}$, $\log 3 = 0.48$, $\log 7 = 0.85$)
- (A) $\text{pH} = 11.44$ (B) $[\text{enH}^+] = 2.7 \times 10^{-3} \text{ M}$
- (C) $[\text{enH}_2^{2+}] = 7.0 \times 10^{-8} \text{ M}$ (D) $[\text{H}^+] = 2.7 \times 10^{-3} \text{ M}$
3. Which of the following graph **does not** represent the radial probability function of 5d electron?

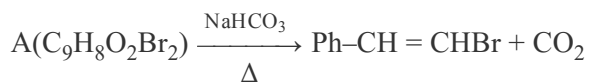


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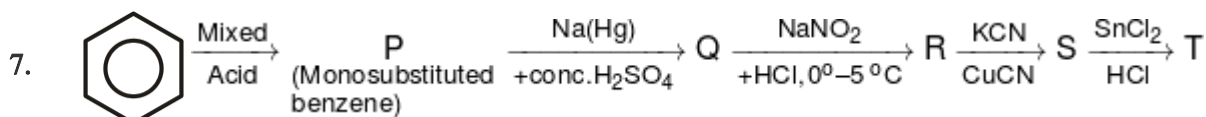
4. In the equilibrium mixture :
 $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, the mole ratio of gases are $\sqrt{2} : \sqrt{2} : 10$, respectively. What would be the effect on the mole ratio on adding 1 mole of He gas at constant pressure?
- (A) No change
 (B) The new molar ratio becomes 1 : 1 : 5
 (C) The new molar ratio becomes 2 : 2 : 5
 (D) The new molar ratio becomes $2 : 2 : 5\sqrt{2}$
5. Which one of the following statement about lactose is **CORRECT** ?
- (A) It does not reduce Tollen's reagent but gives Barfoed's test
 (B) It is a disaccharide. On hydrolysis, it produces two different types of monosaccharides
 (C) It possesses glycosidic linkage
 (D) It shows mutarotation

Space for Rough Work

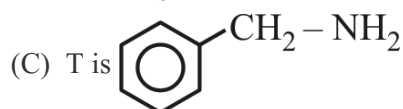
6. Select the **CORRECT** statements about the reaction.



- (A) One possible structure of A have four stereoisomers.
 (B) Two moles of CO_2 are released per mol of A.
 (C) A decolourizes Br_2/H_2O
 (D) One possible structure of A has two stereoisomers

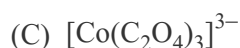


Correct statement are :



(D) P gives positive Mulliken Barker's test

8. Which of the following complexes have higher Δ_0 (splitting energy) than P(pairing energy) :



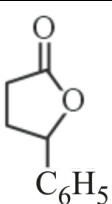
Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Consider the conversion reactions for given organic compounds in List-I. List-II contains appropriate reagents which may be used for effecting these conversions.

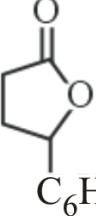
List-I (Conversion)		List-II (Reagent Used)	
(I)	$\text{PhCONH}_2 \longrightarrow \text{PhCH}_2\text{OH}$	(P)	(i) B_2H_6 (ii) $\text{AcOH}/\text{H}_2\text{O}$
(II)	 $\longrightarrow \text{CH}_2\text{OH}-(\text{CH}_2)_2-\text{CH}(\text{OH})-\text{C}_6\text{H}_5$	(Q)	LiAlH_4
(III)	$\text{C}_6\text{H}_5\text{CH}_2-\text{CH}_2-\text{COOH} \longrightarrow \text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$	(R)	$\text{H}_2 / \text{Pd} / \text{BaSO}_4$
(IV)	$\text{CH}_3\text{COCl} \longrightarrow \text{CH}_3-\text{CHO}$	(S)	$\text{NaNO}_2 + \text{HCl}$
		(T)	NaBH_4
		(U)	P.C.C

Which of the following combinations from **List-II** will give **correct** set of reagents to effect proper conversion in **List-I** ?

- (A) III $\rightarrow$ T (B) IV $\rightarrow$ R
 (C) I $\rightarrow$ S (D) IV $\rightarrow$ U

Space for Rough Work

10. Consider the conversion reactions for given organic compounds in List-I. List-II contains appropriate reagents which may be used for effecting these conversions.

List-I (Conversion)		List-II (Reagent Used)	
(I)	$\text{PhCONH}_2 \longrightarrow \text{PhCH}_2\text{OH}$	(P)	(i) B_2H_6 (ii) $\text{AcOH}/\text{H}_2\text{O}$
(II)	 $\rightarrow \text{CH}_2\text{OH}-(\text{CH}_2)_2-\text{CH}(\text{OH})-\text{C}_6\text{H}_5$	(Q)	LiAlH_4
(III)	$\text{C}_6\text{H}_5\text{CH}_2-\text{CH}_2-\text{COOH} \longrightarrow \text{C}_6\text{H}_5-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$	(R)	$\text{H}_2 / \text{Pd} / \text{BaSO}_4$
(IV)	$\text{CH}_3\text{COCl} \longrightarrow \text{CH}_3-\text{CHO}$	(S)	$\text{NaNO}_2 + \text{HCl}$
		(T)	NaBH_4
		(U)	P.C.C

Which of the following is **INCORRECT** combinations?

- (A) $\text{II} \rightarrow \text{Q}$ (B) $\text{IV} \rightarrow \text{R}$
(C) $\text{III} \rightarrow \text{Q}$ (D) $\text{IV} \rightarrow \text{T}$

Space for Rough Work

11. List-I and List-II

List-I		List-II	
(I)	50% tetrahedral void are occupied.	(P)	NaCl
(II)	100% octahedral voids are occupied	(Q)	CsCl
(III)	100% tetrahedral voids are occupied	(R)	Diamond
(IV)	100% cubic voids are occupied	(S)	Sodium oxide
		(T)	Calcium fluoride
		(U)	Zinc wurtzite

 Choose the only **incorrect** combination

- (A) $I \rightarrow R$ (B) $II \rightarrow T$ (C) $III \rightarrow S$ (D) $IV \rightarrow Q$

12. List-I and List-II

List-I		List-II	
(I)	50% tetrahedral void are occupied.	(P)	NaCl
(II)	100% octahedral voids are occupied	(Q)	CsCl
(III)	100% tetrahedral voids are occupied	(R)	Diamond
(IV)	100% cubic voids are occupied	(S)	Sodium oxide
		(T)	Calcium fluoride
		(U)	Zinc wurtzite

 Choose the HCP lattice having **correct** combination

- (A) $II \rightarrow U$ (B) $II \rightarrow P$ (C) $III \rightarrow S$ (D) $I \rightarrow U$

 Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Modern vacuum pumps permit the pressures down to $\frac{1}{\pi} \times 10^{-14}$ bar to be reached at room temperature (27°C). Assuming that the gas exhausted is nitrogen, find the mean free path of gas molecules present at this pressure. [Take diameter of nitrogen molecule as 1.5 Å, $N_A = 6 \times 10^{23}$] In scientific notation, if value is $x \times 10^y$ m then, what is the value of y ?
-

Space for Rough Work

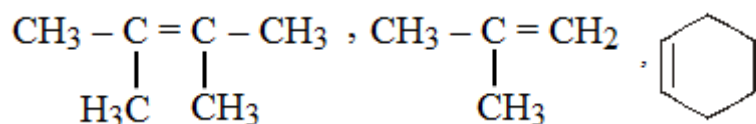
2. (A) The number of carbon atoms in smallest aldose able to form a cyclic stable hemiacetal
 (B) How many phenyl rings are present in major product of the following reaction.



Write the value as a sum of answers of (A), (B)

3. How many out of following compounds have more heat of combustion than $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$?

CH_4 , $\text{CH}_2 = \text{CH}_2$, $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH}_3$, $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH} = \text{CH}_2$, $\text{CH}_2 = \text{CH} - \text{CH}_3$,



Space for Rough Work

4. Write the total number of negatively charged sols among the following :-
Copper sol, CdS sol, Congo red sol, Charcoal sol, Starch sol, TiO_2 sol, Methylene Blue sol, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, Eosin
5. For the reaction, $2\text{A} + \text{B} \rightarrow \text{C} + \text{D}$; rate law is given as $-\frac{d[\text{B}]}{dt} = k[\text{A}]^x [\text{B}]^y$. When $[\text{B}] \gg [\text{A}]$, half life of the reaction was found to be $\frac{1}{7}$ th the time taken for 87.5% completion. However, when $[\text{A}] \gg [\text{B}]$, time taken for 75% completion of reaction was 5 times the half life of the reaction. Find value of $\frac{y}{x}$.
6. In a polymeric structure of Boric acid, each H_3BO_3 molecule is involved in _____ number of hydrogen bonds.
-

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. If f_1, f_2, f_3, f_4 are real functions defined as

$$f_1(x) = \sin^{-1}(\cos(\sin^2 x))$$

$$f_2(x) = \cos^{-1}(\sin(\cos^2 x))$$

$$f_3(x) = \sin^{-1}(\cos(\cos^2 x))$$

$$f_4(x) = \cos^{-1}(\sin(\sin^2 x)),$$

then which of the following option(s) is/are correct :

$$(A) \sum_{i=1}^4 f_i(x) = 2\pi - 2$$

$$(B) \sum_{i=1}^4 (-1)^i f_i(x) = 0$$

$$(C) \sum_{i=1}^4 (-1)^{\left[\frac{i}{3}\right]} f_i(x) = 0 \quad ([.] \rightarrow \text{GIF})$$

$$(D) \text{ If } f_1(x) > f_2(x) \Rightarrow \cos 2x > 0$$

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2. If $A = \sum_{r=1}^3 \log |\tan(60^\circ - \alpha_r)|$, $B = \sum_{r=1}^3 \log |\tan(60^\circ + \alpha_r)|$, where $\alpha_r = \frac{\theta}{3^r}$, then the value of $A+B$ at $\theta = \frac{9\pi}{4}$ is (where base of logarithm is $2 + \sqrt{3}$)
- (A) An even number (B) An odd number
(C) A prime number (D) Neither prime nor composite
3. Let $f(x) = x^2(x^2 + 3x) + x(x - 3x^2) + x(x - 3) - x(4x - 3)$ and $g(x) = \begin{cases} \min\{f(t)\}, & -2 \leq t \leq x, & -2 \leq x < 0 \\ \max\{f(t)\}, & 0 \leq t \leq x, & 0 \leq x \leq 2 \end{cases}$, then
- (A) $g(x)$ is continuous and differentiable at $x = -1$
(B) $g(x)$ is continuous and differentiable at $x = \sqrt{2}$
(C) $g(x)$ is continuous but not differentiable at $x = \sqrt{2}$
(D) $g(x)$ is continuous and differentiable at $x = 1$
4. The value of x in $\left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ can be
- (A) $\frac{\pi}{12}$ (B) $\frac{11\pi}{36}$ (C) $\frac{13\pi}{36}$ (D) $\frac{\pi}{6}$

Space for Rough Work

5. Let $S_n = \sum_{r=1}^n \left(\frac{r^4 + 2nr^3 + r^2n^2 + 3n^4}{n^5} \right)$ and $T_n = \sum_{r=0}^{n-1} \left(\frac{r^4 + 2nr^3 + r^2n^2 + 3n^4}{n^5} \right)$,

where $n = 1, 2, 3, \dots$. Then which of the following is/are correct

(A) $T_n > \frac{121}{30}$

(B) $S_n > \frac{121}{30}$

(C) $T_n < \frac{121}{30}$

(D) $S_n < \frac{121}{30}$

6. If two tangents are drawn from origin to the curve $f(x) = (\ln x)^2 + 2\ln x$, then

(A) Product of abscissae of points of contacts is 1

(B) Product of abscissae of points of contacts is e^2

(C) Number of solution of equation $x (5f'(x) - \ln 10) = 10$ is 1.

(D) Number of solution of equation $x (5f'(x) - \ln 10) = 10$ are 2.

Space for Rough Work

7. If $\vec{a} = \vec{b} + \vec{c}$ and there is a non-zero vector $\vec{d}$ which is parallel to $\vec{b}$ and perpendicular to $\vec{c}$ with $|\vec{c}| = 2$, then which is/are true

(A) $\frac{|\vec{d} \times (\vec{a} \times \vec{d})|}{|\vec{d}|^2} = 1$

(B) $\frac{|\vec{d} \times (\vec{a} \times \vec{d})|}{|\vec{d}|^2} = 2$

(C) $\frac{|\vec{b} \cdot \vec{a}|}{|\vec{b} \cdot \vec{d}|} = \frac{|\vec{b}|}{|\vec{d}|}$

(D) $\frac{|\vec{b} \cdot \vec{a}|}{|\vec{b} \cdot \vec{d}|} = \frac{|\vec{d}|}{|\vec{b}|}$

8. Let $A_n = \begin{bmatrix} x_n & 0 & 0 \\ 0 & x_{n+1} & 0 \\ 0 & 0 & 1 \end{bmatrix}$. The sequence of terms $x_1, x_2, x_3, \dots$ satisfying $x_1 = 2, x_2 = 5$ and

$x_{i+2} = \frac{5}{x_{i+1}} + x_i, \forall i = 1, 2, 3, \dots$. If the value of $\det(A_{150})$ is k , then

- (A) k is divisible by 10
(B) sum of digits of k is prime
(C) sum of digit of k is divisible by 5
(D) k is divisible by 17

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Each List-Match set has **TWO** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
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 Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered)
 Negative Marks : -1 In all other cases

9. Let $f : A \rightarrow B$, $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4\}$. List-I contains condition on function and List-II contains corresponding number of function.

List-I		List-II	
(I)	The number of onto function of f	(P)	35
(II)	The number of function f such that $f(i) \leq f(j)$, $\forall i < j$ and $i, j \in \{1, 2, \dots, 5\}$	(Q)	570
(III)	The number of function of f such that $\{1, 2\} \subseteq \text{Range of } f$	(R)	572
(IV)	The number of function of f such that $f(1) = 4$ and $f(i) \geq f(j)$, $\forall i < j$ and $i, j \in \{2, 3, 4, 5\}$	(S)	240
		(T)	244
		(U)	56

Which of the following is only correct combination ?

- (A) $I \rightarrow Q; II \rightarrow U$ (B) $I \rightarrow S; II \rightarrow Q$ (C) $I \rightarrow R; II \rightarrow U$ (D) $I \rightarrow S; II \rightarrow U$
10. Let $f : A \rightarrow B$, $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4\}$. List-I contains condition on function and List-II contains corresponding number of function.

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(II)	The number of function f such that $f(i) \leq f(j)$, $\forall i < j$ and $i, j \in \{1, 2, \dots, 5\}$	(Q)	570
(III)	The number of function of f such that $\{1, 2\} \subseteq \text{Range of } f$	(R)	572
(IV)	The number of function of f such that $f(1) = 4$ and $f(i) \geq f(j)$, $\forall i < j$ and $i, j \in \{2, 3, 4, 5\}$	(S)	240
		(T)	244
		(U)	56

Which of the following is only correct combination ?

- (A) $III \rightarrow Q; IV \rightarrow S$ (B) $III \rightarrow S; IV \rightarrow P$ (C) $III \rightarrow Q; IV \rightarrow P$ (D) $III \rightarrow S; IV \rightarrow U$

11. A box contains 25 tickets each with a different number from 1 to 25. Four tickets are drawn at random without replacement. Let a, b, c, d be the numbers of the tickets drawn. List-I contain some restricted values of a, b, c, d and List-II contain corresponding probabilities.

List-I		List-II	
(I)	The probability that $\min(a, b, c, d) < 10$ is equal to :	(P)	$\frac{189}{2530}$
(II)	The probability that 10 is the second smallest number is	(Q)	$\frac{286}{575}$
(III)	The probability that the sum of $a + b + c + d$ is odd is	(R)	$\frac{207}{230}$
(IV)	The probability that the product of $abcd$ is even is	(S)	$\frac{217}{230}$
		(T)	$\frac{1083}{1265}$
		(U)	$\frac{1083}{1256}$

Which of the following is only correct combination ?

- (A) $I \rightarrow Q; III \rightarrow S$ (B) $I \rightarrow T; III \rightarrow Q$
 (C) $I \rightarrow Q; III \rightarrow R$ (D) $I \rightarrow T; III \rightarrow S$
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(III)	The probability that the sum of $a + b + c + d$ is odd is	(R)	$\frac{207}{230}$
(IV)	The probability that the product of $abcd$ is even is	(S)	$\frac{217}{230}$
		(T)	$\frac{1083}{1265}$
		(U)	$\frac{1083}{1256}$

Which of the following is only correct combination ?

- (A) $II \rightarrow P; IV \rightarrow R$ (B) $II \rightarrow U; IV \rightarrow S$
 (C) $II \rightarrow P; IV \rightarrow S$ (D) $II \rightarrow R; IV \rightarrow S$

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Let $A = [a_{ij}]_{3 \times 3}$ and $a_{ij} \neq 0$, be a matrix such that $AA' = 4I$ and $2a_{ij} + c_{ij} = 0 \forall i, j$, where c_{ij} is co-factor of a_{ij} and I is unit matrix of order 3,
$$\begin{vmatrix} a_{11} + 4 & a_{12} & a_{13} \\ a_{21} & a_{22} + 4 & a_{23} \\ a_{31} & a_{32} & a_{33} + 4 \end{vmatrix} + 5\lambda \begin{vmatrix} a_{11} + 1 & a_{12} & a_{13} \\ a_{21} & a_{22} + 1 & a_{23} \\ a_{31} & a_{32} & a_{33} + 1 \end{vmatrix} = 0$$
 then λ equals (where A' denotes transpose of matrix A)
2. Let $a, b, c > 0$ and $\frac{1}{1+a^4} + \frac{1}{1+b^4} + \frac{1}{1+c^4} = 1$, then minimum value of $a^2b^2c^2$. (use $\sqrt{2} = 1.41$ and $\sqrt{3} = 1.73$)
3. Let $f(x)$ be a polynomial with integral coefficient and has atleast one integral root. If $f(2) = 13$ and $f(10) = 5$, then the value of α such that $f(\alpha) = 0$ (where $\alpha \in I$)
-

Space for Rough Work

4. Positive number a, b and c satisfy $ab^2c^3 = (1024) \cdot 2^{71}$ and $(2\log_2 a)(\log_2 bc^{3/2}) + 6\log_2 b \cdot \log_2 c = 468$.
The value of $\sqrt{(\log_2 a)^2 + (2\log_2 b)^2 + (3\log_2 c)^2}$ is
5. If the possible solution of the differential equation $y(y^2 - 2x^2) \cdot dx + x \cdot (2y^2 - x^2) \cdot dy = 0$ is of the form $f(x) \cdot g(y) \cdot (g(y) - f(x)) = C$ where $f(x), g(y)$ are monic polynomials and C arbitrary constant then the value of $\frac{g(3)}{f(2)}$ is
6. If planes, $P_1 : x = y \sin \alpha + z \sin \beta$, $P_2 : y = x \cdot \sin \alpha + z \cdot \sin \theta$, $P_3 : z = x \sin \beta + y \sin \theta$ pass through the line $\frac{x}{\cos \theta} = \frac{y}{\cos \beta} = \frac{z}{\cos \alpha}$, then sum of all possible values of $\alpha + \beta + \theta$. where $\alpha + \beta + \theta \in [0, 6\pi]$ (take $\pi = 3.14$)

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work